

Basic Gradient Boosting Regressor Algorithm:

Highest R2_Score: 0.9248821037

S.NO	loss	criterion	max_features	R2_score
1	squared_error	friedman_mse	None	0.920846743
2		friedman_mse	sqrt	0.7849795061
3		friedman_mse	log2	0.8135087488
4		squared_error	None	0.9248821037
5		squared_error	sqrt	0.7582363112
6		squared_error	log2	0.7712128003
7	absolute_error	friedman_mse	None	0.8529747106
8		friedman_mse	sqrt	0.7063681242
9		friedman_mse	log2	0.6672040775
10		squared_error	None	0.8814781692
11		squared_error	sqrt	0.7900856689
12		squared_error	log2	0.646852447
13	huber	friedman_mse	None	0.9245386576
14		friedman_mse	sqrt	0.7863043503
15		friedman_mse	log2	0.7843580111
16		squared_error	None	0.9244958052
17		squared_error	sqrt	0.7877686897
18		squared_error	log2	0.7877825253
19	quantile	friedman_mse	None	0.8566235546
20		friedman_mse	sqrt	0.7619887909

21		friedman_mse	log2	0.7725127079
22		squared_error	None	0.7974496765
23		squared_error	sqrt	0.7425773334
24		squared_error	log2	0.713663042

Ada Boosting Algorithm:

Highest R2_score : 0.9340036334

<u>S.No</u>	loss	R2_Score
1	linear	0.9340036334
2	square	0.9242184418
3	exponential	0.9275843712

XG Boosting Algorithm

Highest r2_score: 0.9536390305

<u>S.No</u>	objective	tree_method	booster	R2_Score
1	reg:squarederror'	auto	gbtree	0.9274241328
2		auto	gblinear	0.9536023736
3		auto	dart	0.9274240136
4		hist	gbtree	0.9274241328
5		hist	gblinear	0.953514576

6		hist	dart	0.9274240136
7		exact	gbtree	0.9003389478
8		exact	gblinear	0.9536390305
9		exact	dart	0.9003391266
10	'reg:absoluteerror'	auto	gbtree	0.8723513484
11		auto	gblinear	-0.05674743652
12		auto	dart	0.8723515868
13		hist	gbtree	0.8723513484
14		hist	gblinear	-0.05675625801
15		hist	dart	0.8723515868
16		exact	gbtree	-0.05616044998
17		exact	gblinear	-0.05674064159
18		exact	dart	-0.05616044998

LG boosting Algorithm

Highest R2_score= -0.03677248112

S.No	objective	boosting_type	r2Score
1	regression	gbdt	-0.03677248112
2		dart	-0.03677248112
4	mae	gbdt	-0.05748625661
5		dart	-0.05748625661
7	huber	gbdt	0.03677248112

8		dart	-0.03677248112
---	--	------	----------------